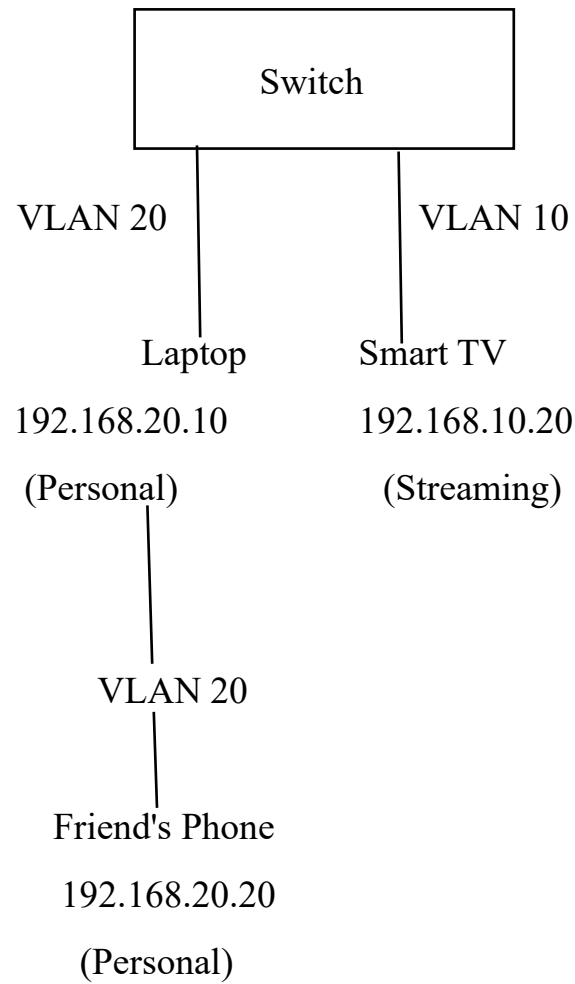


SWITCHING AND VLANS

1. Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).



Device	VLAN	Purpose
Laptop	VLAN 20	Personal Devices
Friend's Phone	VLAN 20	Personal Devices

Device	VLAN	Purpose
Smart TV	VLAN 10	Streaming Devices

**2. Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.

Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.**

- **Create VLANs**
enable
configure terminal
vlan 10
vlan 20
- **Assign PCs to VLAN 10 (PC1–PC3)**
interface range fa0/1-3
switchport mode access
switchport access vlan 10
exit
- **Assign PCs to VLAN 20 (PC4–PC5)**
interface range fa0/4-5
switchport mode access
switchport access vlan 20
end

IP Addresses

- VLAN 10: PC1 (192.168.10.10), PC2 (192.168.10.11), PC3 (192.168.10.12)

- VLAN 20: PC4 (192.168.20.10), PC5 (192.168.20.11)

Verification

- PC1 → PC2: Ping successful (same VLAN).
- PC1 → PC4: Ping failed (different VLANs).

Conclusion: Devices in the same VLAN can communicate, while devices in different VLANs cannot without inter-VLAN routing.

3. In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

- **Before (PC1 in VLAN 10)**

enable

configure terminal

interface fa0/1

switchport mode access

switchport access vlan 10

Test:

- PC1 → PC2 (VLAN 10): Ping Success
- PC1 → PC4 (VLAN 20): Ping Failed

- **After (Move PC1 to VLAN 20)**

enable

configure terminal

interface fa0/1

switchport mode access

```
switchport access vlan 20
```

Test:

- PC1 → PC2 (VLAN 10): Ping Failed
- PC1 → PC4 (VLAN 20): Ping Success

Conclusion

After moving PC1 from VLAN 10 to VLAN 20, it can communicate with devices in VLAN 20 but cannot communicate with devices in VLAN 10.

4. Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).
Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

Cisco Packet Tracer / NetSim Command List

```
enable  
configure terminal  
vlan 30  
name VLAN30  
interface range fa0/1-10  
switchport mode access  
switchport access vlan 30  
end  
write memory
```

Explanation

- vlan 30 → Creates VLAN 30.

- interface range fa0/1-10 → Selects ports FastEthernet 0/1 to 0/10 at once.
- switchport mode access → Sets the ports as access ports.
- switchport access vlan 30 → Assigns all 10 ports to VLAN 30.
- write memory → Saves the configuration.

This uses the interface range command, so you don't have to configure each port individually.